Docker image deployed:

```
joaoribeiro@JOCAPC:~/java-k8s-lab$ docker ps
CONTAINER ID   IMAGE                                   COMMAND                  CREATED          STATUS          P
ORTS
                     NAMES
78cf1faee11f   my-java-app:1.0                         "java -jar app.jar"      56 seconds ago   Up 56 seconds   0
.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
                     test-java-app
0a759f404b06   gcr.io/k8s-minikube/kicbase:v0.0.50   "/usr/local/bin/entr…"   3 days ago       Up 4 hours      1
27.0.0.1:32768->22/tcp, 127.0.0.1:32769->2376/tcp, 127.0.0.1:32770->5000/tcp, 127.0.0.1:32771->8443/tcp, 127.0.0
.1:32772->32443/tcp   minikube
```

Docker image pushed to my dockerHub

```
joaoribeiro@JOCAPC:~/java-k8s-lab$ docker tag my-java-app:1.0 joaoribeiro9595/my-java-app:1.0
joaoribeiro@JOCAPC:~/java-k8s-lab$ docker push joaoribeiro9595/my-java-app:1.0
The push refers to repository [docker.io/joaoribeiro9595/my-java-app]
d5b5ee339aff: Pushed
1a8b5eadd33c: Pushed
83b767b06655: Mounted from library/adoptopenjdk
14fbd8039ba4: Mounted from library/adoptopenjdk
da55b45d310b: Mounted from library/adoptopenjdk
1.0: digest: sha256:31ed1860e2811874b303192fae7afe7adc1b0d52ee30465827d0730b9571b93d size: 1367
```

Java App:

HelloController.java

```java
import com.sun.net.httpserver.HttpServer;
import com.sun.net.httpserver.HttpHandler;
import com.sun.net.httpserver.HttpExchange;
import java.io.IOException;
import java.io.OutputStream;
import java.net.InetSocketAddress;

public class HelloController {
    public static void main(String[] args) throws IOException {
        HttpServer server = HttpServer.create(new InetSocketAddress(8080), 0);
        server.createContext("/", new MyHandler());
        server.setExecutor(null);
        server.start();
        System.out.println("Server started on port 8080...");
    }

    static class MyHandler implements HttpHandler {
        @Override
        public void handle(HttpExchange t) throws IOException {
            String response = "Hello, Kubernetes!";
            t.sendResponseHeaders(200, response.length());
            try (OutputStream os = t.getResponseBody()) {
                os.write(response.getBytes());
            }
        }
    }
```

Dockerfile for the Java App:

```dockerfile
# --------- Build Stage ---------
FROM adoptopenjdk:11 as build
WORKDIR /app

# Copy the Java source file into the container
COPY HelloController.java .

# Compile the Java source file
RUN javac HelloController.java

# Package all compiled classes (including inner classes) into an executable jar
RUN jar cfe app.jar HelloController *.class

# --------- Run Stage ---------
FROM adoptopenjdk:11-jre-hotspot
WORKDIR /app

# Copy the executable jar from the build stage
COPY --from=build /app/app.jar .

# Expose the port on which the app listens
EXPOSE 8080

# Run the jar file
ENTRYPOINT ["java", "-jar", "app.jar"]
```

java-deployment:

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: java-app-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: java-app
  template:
    metadata:
      labels:
        app: java-app
    spec:
      containers:
      - name: java-app-container
        image: joaoribeiro9595/my-java-app:1.0
        ports:
        - containerPort: 8080
```

Java-service:

```yaml
apiVersion: v1
kind: Service
metadata:
  name: java-app-service
spec:
  type: NodePort
  selector:
    app: java-app
  ports:
    - protocol: TCP
      port: 80          # The service port
      targetPort: 8080 # Container's port
      nodePort: 30080  # Node port (optional)
```

Searching for our pod:

```
joaoribeiro@JOCAPC:~/java-k8s-lab$ kubectl get pods
NAME                                  READY   STATUS    RESTARTS        AGE
backend-66569f457c-h9g2j              1/1     Running   4 (43m ago)     3d2h
backend-66569f457c-k94hh              1/1     Running   4 (43m ago)     3d2h
backend-66569f457c-mpdpv              1/1     Running   4 (43m ago)     3d2h
frontend-6fcd66b77d-qr8f5             1/1     Running   14 (43m ago)    3d2h
java-app-deployment-74f9d95cc4-zrpwv  1/1     Running   0               3m12s
secret-configmap-pod                  1/1     Running   1 (43m ago)     3h39m
```

Searching for our deployment:

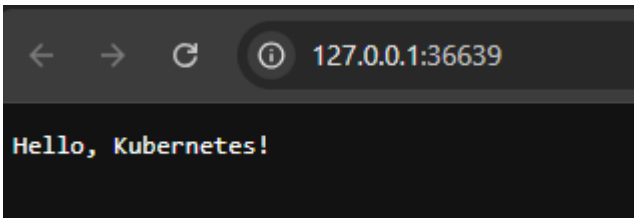
```
joaoribeiro@JOCAPC:~/java-k8s-lab$ kubectl get deployment
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
backend               3/3     3            3           3d2h
frontend              1/1     1            1           3d2h
java-app-deployment   1/1     1            1           18s
redis-deployment      1/1     1            1           39m
```

Searching for our service:

```
joaoribeiro@JOCAPC:~/java-k8s-lab$ kubectl get svc
NAME               TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
backend-service    ClusterIP   10.106.148.23   <none>        3000/TCP       3d2h
frontend-service   NodePort    10.99.175.111   <none>        80:32130/TCP   3d2h
java-app-service   NodePort    10.97.232.41    <none>        80:30080/TCP   19s
kubernetes         ClusterIP   10.96.0.1       <none>        443/TCP        3d5h
```

Launching the Java App:

```
🔗 Starting tunnel for service java-app-service.
```

NAMESPACE	NAME	TARGET PORT	URL
default	java-app-service		http://127.0.0.1:36639

Browser connected and responding: